

Midterm – MATH 346/377 – Winter 2025

DATE: Wednesday, February 26

TIME: 8:35 – 9:55

1. (10 points) True or false? Justify your answer.

- (a) Let a, b, c be nonzero integers. Then $\gcd(a, b) + c = \gcd(a + c, b + c)$.
- (b) The equation $319x + 30y = n$ has a solution $x, y \in \mathbb{Z}$ for any n .
- (c) There are four elements of order 5 modulo 13.
- (d) If $\gcd(n, m) = 1$, then $\gcd(\phi(n), \phi(m)) = 1$.

Solution:

- (a) FALSE. Take $a = 3$, $b = 5$ and $c = 2$. Then $\gcd(a + c, b + c) = \gcd(5, 7) = 1$ but $\gcd(a, b) + c = \gcd(3, 5) + 2 = 3$.
- (b) TRUE. By Bezout's theorem, the equation $319x + 30y = n$ has a solution iff $\gcd(319, 30)$ divides n . By Euclidean division, we find that $\gcd(319, 30) = \gcd(30, 19) = \gcd(19, 11) = 1$. Hence the equation has a solution for any n .
- (c) FALSE. The order modulo 13 of any element must divide $\phi(13) = 12$. Since 5 is coprime to 13, there are no elements of order 5.
- (d) FALSE. For any $n > 2$ we have that $\phi(n)$ is even. So for example take the coprime integers $n = 3$ and $m = 5$, then $\phi(n) = 2$ and $\phi(m) = 4$ are not coprime.

2. (BONUS 10 points) Let $a \geq 1$ and $n \geq 2$ be two integers.

- (a) (5 points) If $a^n - 1$ is prime, show that $a = 2$.
- (b) (5 points) If $2^n - 1$ is prime, show that n is prime.

Primes of the form $2^p - 1$ are called Mersenne primes, and are not related to Fermat primes $2^{2^n} + 1$.

Solution:

- (a) We know that

$$a^n - 1 = (a - 1)(1 + a + \cdots + a^{n-1}). \quad (1)$$

If $a^n - 1$ is prime, we must have $a - 1 = 1$ or $1 + a + \cdots + a^{n-1} = 1$. The latter cannot be true, since $n \geq 2$ and $a \geq 1$. Hence, we must have $a - 1 = 1$, meaning $a = 2$.

(b) We show the contraposition. Suppose that n is not prime, say $n = kq$ with $k, q > 1$. Recall that

$$x^k - 1 = (x - 1)(1 + x + \cdots + x^{k-1}) \quad (2)$$

If we plug-in $x = 2^q$ we see that $2^q - 1$ divides $2^n - 1 = 2^{kq} - 1$. Hence $2^n - 1$ is not prime.

3. (10 points) Find all integers $0 \leq x \leq 54$ that solve the equation

$$f(x) = 3x^{22} + 11x^5 + 8x^2 + 2x - 5 \equiv 0 \pmod{55}.$$

Solution: By the Chinese Remainder Theorem, since 5 and 11 are coprime we have $f(x) \equiv 0 \pmod{55}$ iff $f(x) \equiv 0 \pmod{5}$ and $f(x) \equiv 0 \pmod{11}$.

First we look at the equation modulo 11. For any x we have $x^{11} \equiv x \pmod{11}$ by Fermat's little theorem, hence $x^{22} \equiv x^{2 \cdot 11} \equiv x^2 \pmod{11}$. It follows that

$$f(x) \equiv 3x^2 + 8x^2 + 2x - 5 \equiv 2x - 5 \pmod{11}. \quad (3)$$

So $2x \equiv 5 \pmod{11}$, which implies

$$x \cdot \equiv 52^{-1} \equiv 2 \cdot 6 \equiv 30 \equiv 8 \pmod{11}. \quad (4)$$

Now we consider the equation modulo 5. By Fermat's little theorem, for any x we have $x^5 \equiv x \pmod{5}$ so that $x^{22} \equiv (x^5)^4 x^2 \equiv x^6 \equiv x^2 \pmod{5}$. Thus, we have

$$f(x) = 3x^{22} + 11x^5 + 8x^2 + 2x - 5 \equiv 3x^2 + x + 3x^2 + 2x \equiv x^2 + 3x = x(x + 3) \pmod{5}.$$

We find that $x \equiv 0 \pmod{5}$ or $x \equiv 2 \pmod{5}$.

There are five integers $0 \leq x \leq 54$ congruent to 8 modulo 11:

$$8, 19, 30, 41, 52. \quad (5)$$

Among these five there are two integers that are coprime to 0 or 2 modulo 5:

$$30, 52. \quad (6)$$

So the solutions are $\{30, 52\}$.

4. (10 points) (a) (3 points) How many distinct primitive roots are there modulo 19?
 (b) (7 points) Let g be a primitive root modulo 19. Show that $g^6 \equiv 7 \pmod{19}$ or $g^6 \equiv 11 \pmod{19}$.

Solution:

(a) There are $\phi(\phi(19)) = \phi(18) = \phi(2)\phi(3^2) = 6$ primitive roots modulo 19.

(b) Let g be a primitive root modulo 19, and let $a := g^6$. Then $a^3 \equiv g^{18} \equiv 1 \pmod{19}$. So a is a root of

$$x^3 - 1 = (x - 1)(1 + x + x^2). \quad (7)$$

Since g is a primitive root, $a \not\equiv 1$ and so a must be a root of $x^2 + x + 1 \pmod{19}$. The discriminant is $\Delta = 1 - 4 = -3 \equiv 16 = 4^2 \pmod{19}$, so $x^2 + x + 1$ has two solutions modulo 19:

$$x \equiv \frac{-1 \pm 4}{2} \pmod{19}. \quad (8)$$

The inverse of 2 modulo 19 is 10, so we find the two solutions $x \equiv 10(-5) \equiv -50 \equiv -12 \equiv 7 \pmod{19}$ and $x \equiv 10 \cdot 3 \equiv 30 \equiv 11 \pmod{19}$.

ALTERNATIVE SOLUTION: $x = g^6$ is of order 3, since $x \neq 1$ and $x^2 = g^{12} \neq 1$. Since 3 divides 18, there are $\phi(3) = 2$ elements of order 3. We can easily verify that 7 and 11 are two distinct elements of order 3, so x must be 7 or 11.

5. (BONUS 10 points) Is the number

$$\alpha := \frac{1}{\sqrt{\sqrt{2} + \sqrt{3}}}$$

an algebraic *integer*? Is it a rational number?

Solution: Let $\alpha := \sqrt{\sqrt{2} + \sqrt{3}}$ and

$$\beta = \alpha^2 = \sqrt{2} + \sqrt{3}. \quad (9)$$

We have

$$\beta^2 = 5 + 2\sqrt{6} \quad (10)$$

and

$$\beta^4 = (5 + 2\sqrt{6})^2 = 49 + 20\sqrt{6} = 49 + 10(\beta^2 - 5) = -10\beta^2 - 1. \quad (11)$$

So $\beta^4 + 10\beta^2 + 1$ and α is a root of

$$f(x) = x^8 + 10x^4 + 1. \quad (12)$$

Now $1/\alpha$ is a root of the polynomial,

$$g(x) = x^8 f(1/x) = x^8 + 10x^4 + 1, \quad (13)$$

which is monic and has integer coefficients, so $1/\alpha$ is an algebraic integer. By the rational root theorem, any rational root of $g(x)$ must be a divisor of the constant term 1. So the only possible rational roots of $g(x)$ are 1 and -1 . Since $g(\pm 1) = 12 \neq 0$, the polynomial $g(x)$ has no rational roots and α is irrational.